Individuals Post-Stroke Change Energetics of Multiple Lower Limb Constituents During Gait While Wearing a Passive-Dynamic Ankle Foot Orthosis

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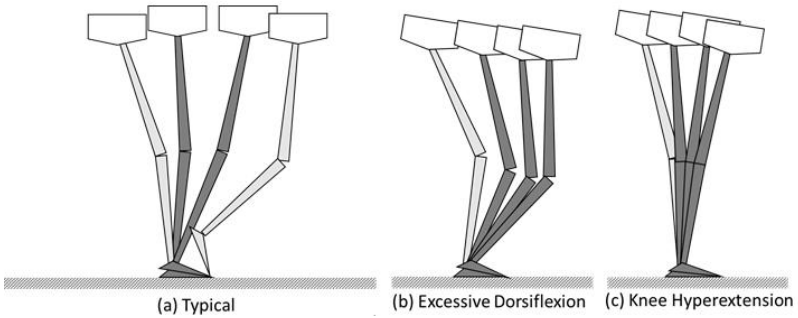
Background

- 795,000 strokes in the United States every year[1]
- PF weakness common from strokes causes walking difficulties



- Passive-dynamic ankle-foot orthoses (PD-AFOs) have been shown to assist weakened plantar flexors[2]

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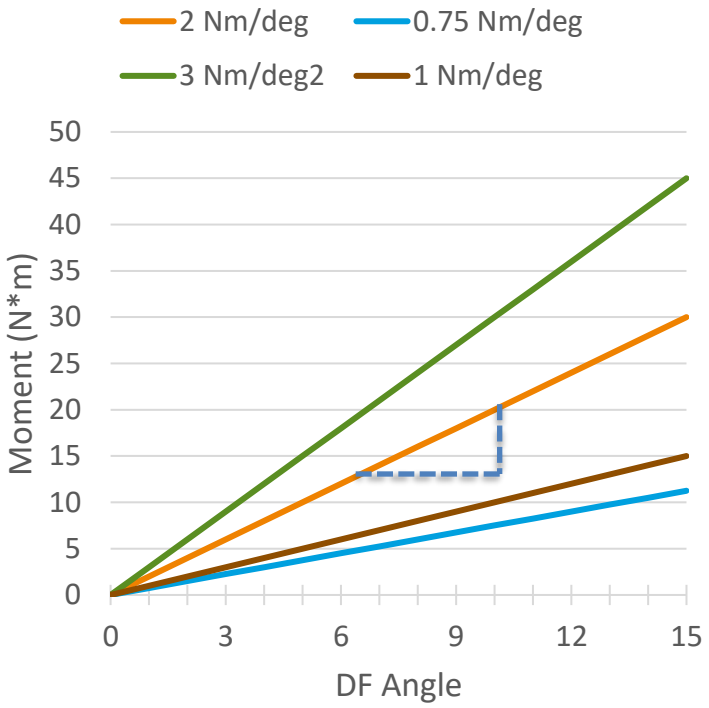
1. cdc.gov/stroke/index.htm
2. Koller, C. et al. Pros Orth Intl, 45, 313-321, 2021.(a)

PD-AFOs

- Plantar flexors act like rubber bands to control ankle position during stance (kinematic) and to store and return energy (kinetic)
- PD-AFO stiffness supplements weakened plantar flexors

- Stiffness = resistance to bending
  - Moment increases with dorsiflexion like a torsion spring
  - Varies between individuals



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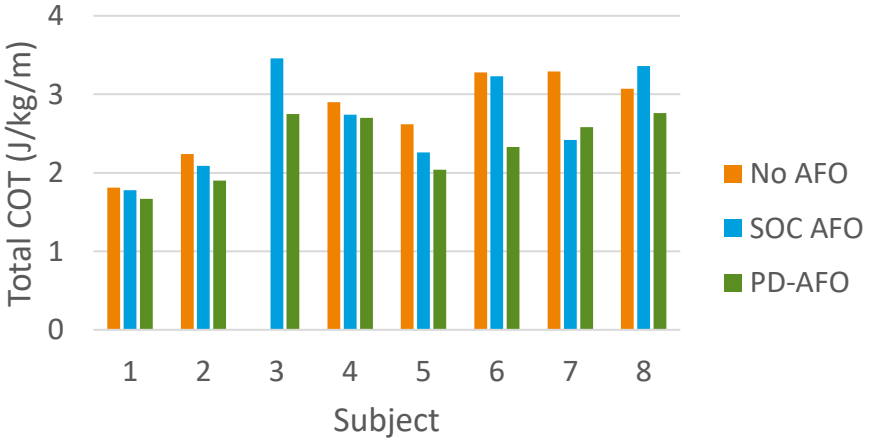
Prior Work

- Stiffness-customized PD-AFOs reduce mechanical cost of transport (COT) compared to walking with either No AFO or standard of care (SOC) AFOs[3]



- **We do not know the drivers behind COT reduction**

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3. Koller, C. et al. Proc. ASB Annual Meeting, 2021.(b)

Purpose:

Investigate how individuals achieve COT reduction while wearing a stiffness-customized PD-AFO

Hypothesis:

COT reduction is driven primarily by decreased paretic ankle work while wearing a PD-AFO compared to No AFO and SOC AFO

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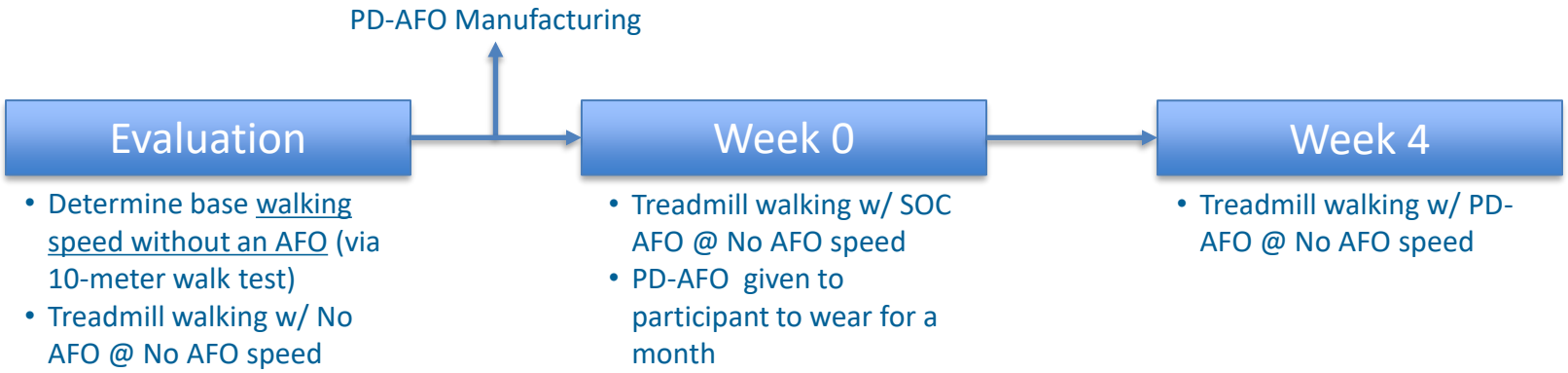
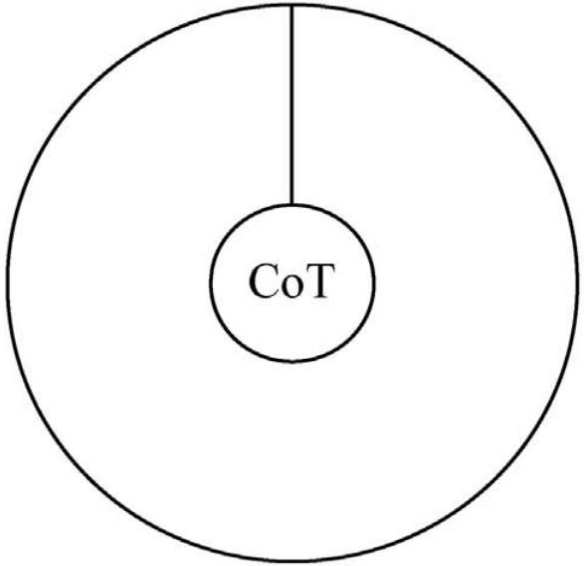
Methods

- 8 individuals (3 male, 5 female) with single unilateral chronic stroke
  - Age 65.1 (6.9) years; Height 1.72 (0.13) m; Mass 82.4 (14.8) kg



- PD-AFO stiffness based on participants' level of plantar-flexor weakness

Constituent Lower Extremity Work (CLEW)



$$COT = \frac{\frac{hip^{+} + knee^{+} + ankle^{+} + foot^{+}}{mass} + abs\left(\frac{hip^{-} + knee^{-} + ankle^{-} + foot^{-}}{mass}\right)}{stride\ length} = \frac{J/kg}{m}$$

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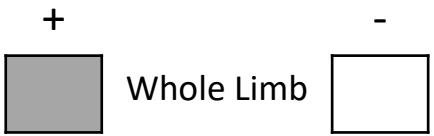
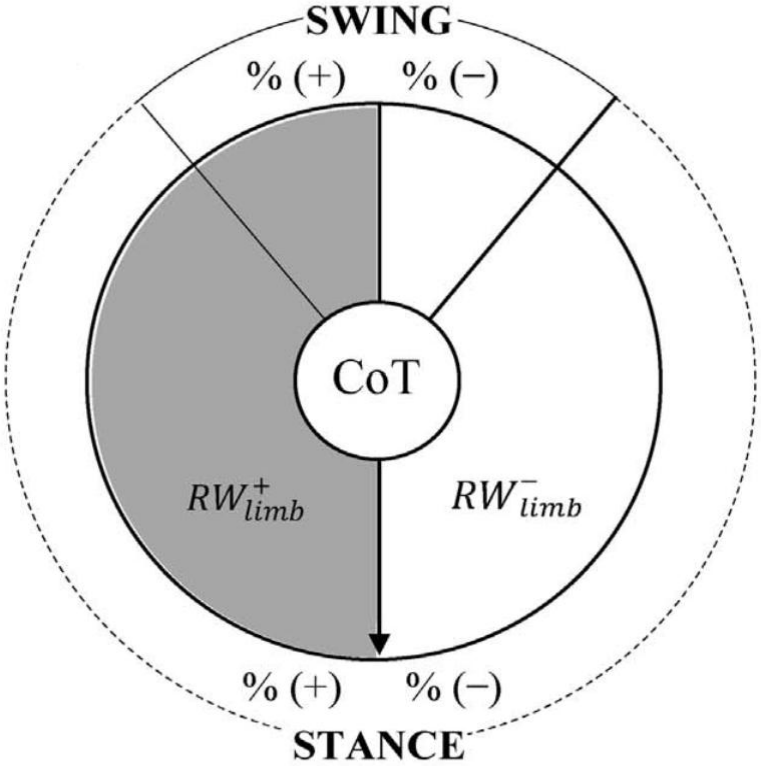
4. Ebrahimi, A. et al. Gait Posture, 56, 49-53, 2017.

Constituent Lower Extremity Work (CLEW)





$$COT = \frac{J/kg}{m}$$

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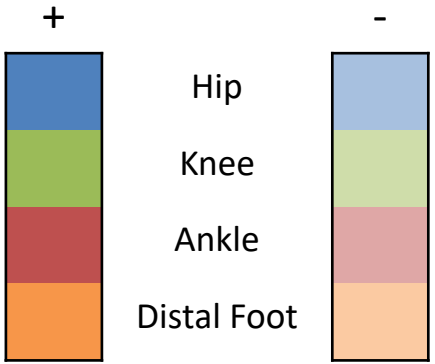
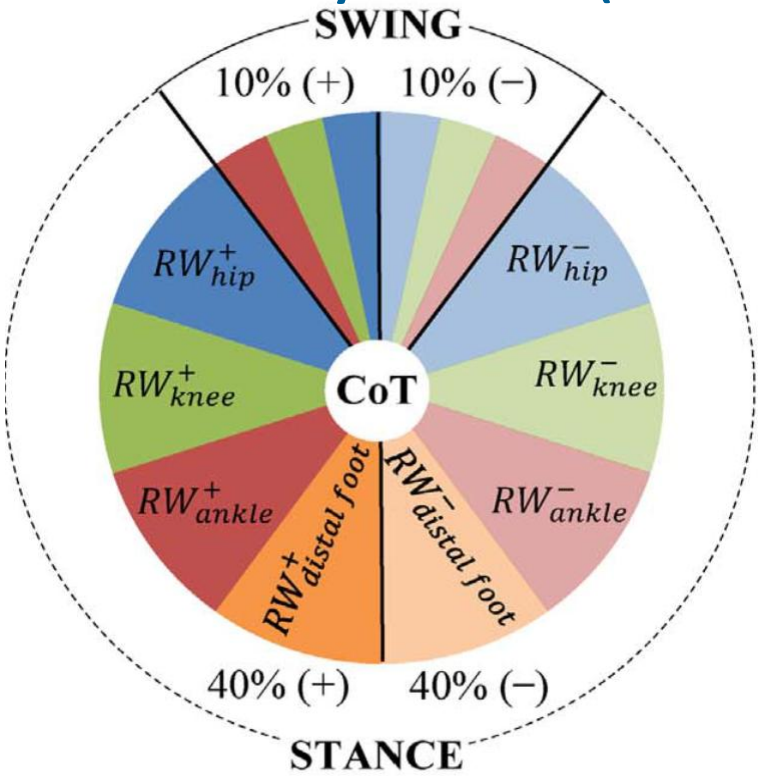
Constituent Lower Extremity Work (CLEW)





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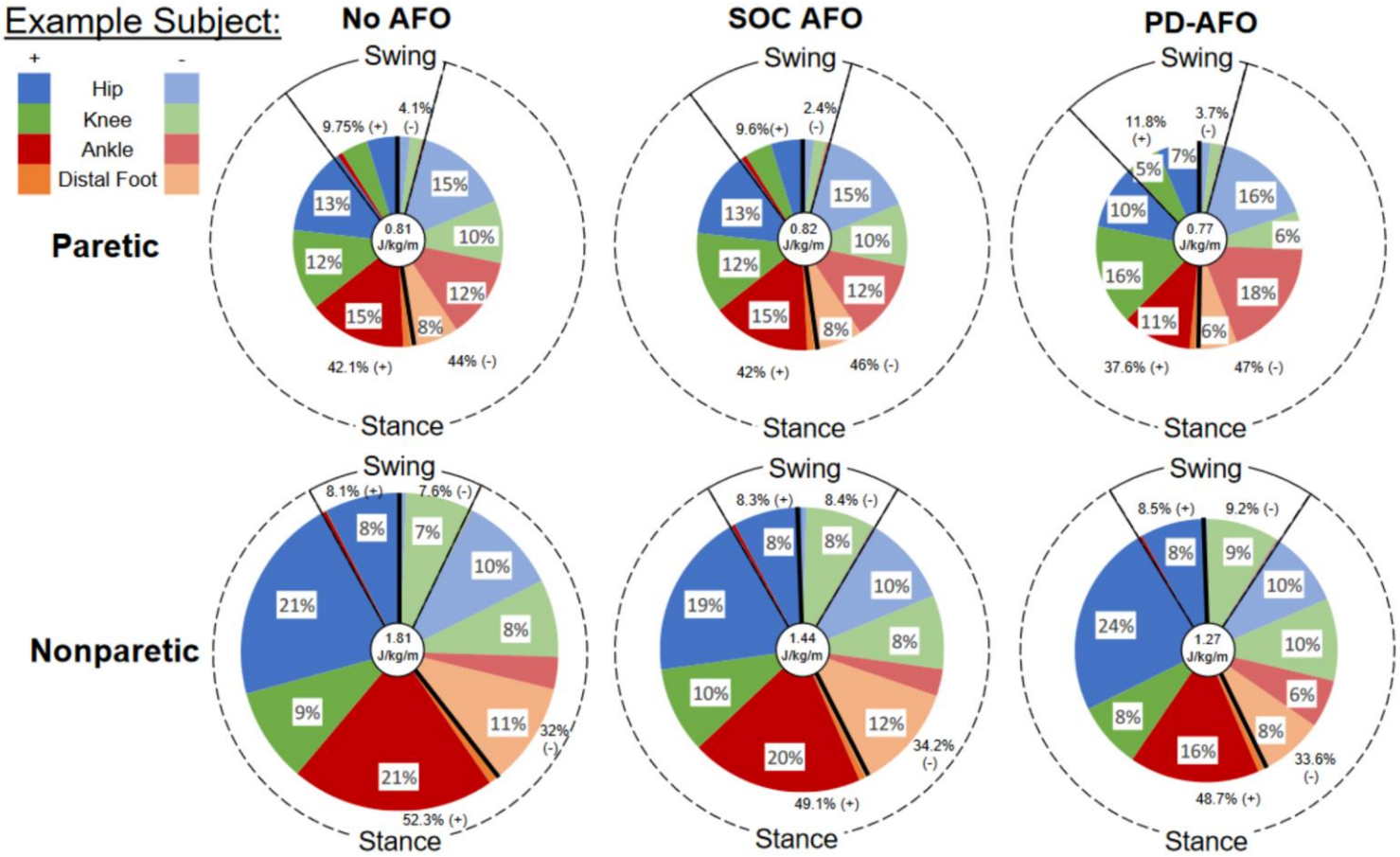
Example Subject:
No AFO
SOC AFO
PD-AFO
+
-
Hip
Knee
Ankle
Distal Foot
Paretic
Nonparetic
Swing
Stance
0.81 J/kg/m
9.75% (+)
4.1% (-)
15%
10%
12%
8%
44% (-)
15%
12%
13%
42.1% (+)
0.82 J/kg/m
9.6%(+)
2.4% (-)
15%
10%
12%
8%
46% (-)
15%
12%
13%
42% (+)
0.77 J/kg/m
11.8% (+)
3.7% (-)
7%
16%
6%
18%
6%
47% (-)
11%
16%
10%
5%
37.6% (+)
1.81 J/kg/m
8.1% (+)
7.6% (-)
8%
7%
10%
8%
11%
32% (-)
21%
52.3% (+)
21%
9%
21%
1.44 J/kg/m
8.3% (+)
8.4% (-)
8%
8%
10%
8%
12%
34.2% (-)
19%
10%
20%
49.1% (+)
1.27 J/kg/m
8.5% (+)
9.2% (-)
8%
9%
10%
10%
6%
8%
33.6% (-)
24%
8%
16%
48.7% (+)

Results

- ⬇ Total COT with the PD-AFO compared to No AFO **and** SOC AFO

Subject	No AFO vs PD-AFO		SOC AFO vs PD-AFO	
	Paretic	Nonparetic	Paretic	Nonparetic
	ΔCOT (J/kg/m)	*ΔCOT (J/kg/m)*	*ΔCOT (J/kg/m)*	*ΔCOT (J/kg/m)*
P1	-0.01	-0.13	-0.01	-0.1
P2	-0.05	-0.29	0.12	-0.31
P3	-±	-±	-0.55	-0.16
P4	0	-0.2	-0.06	0.02
P5	-0.04	-0.54	-0.05	-0.17
P6	-0.55	-0.4	-0.32	-0.58
P7	-0.54	-0.17	-0.03	0.19
P8	-0.27	-0.04	-0.59	-0.01
Mean **SD**	**-0.21** **0.25**	**-0.25** **0.17**	**-0.19** **0.27**	**-0.14** **0.23**

Results

- ⬇ Total COT with the PD-AFO compared to No AFO **and** SOC AFO
- Small changes in **numerous** constituents
- Exact constituents varied across subjects

Subject	No AFO vs PD-AFO				SOC AFO vs PD-AFO			
	Paretic		Nonparetic		Paretic		Nonparetic	
	#	ΔCOT (J/kg/m)	#	ΔCOT (J/kg/m)	#	ΔCOT (J/kg/m)	#	ΔCOT (J/kg/m)
P1	5	-0.01	11	-0.13	3	-0.01	6	-0.1
P2	4	-0.05	11	-0.29	5	0.12	8	-0.31
P3	12±	-±	-±	-±	11	-0.55	12	-0.16
P4	10	0	13	-0.2	9	-0.06	7	0.02
P5	14	-0.04	12	-0.54	14	-0.05	10	-0.17
P6	8	-0.55	7	-0.4	9	-0.32	6	-0.58
P7	7	-0.54	5	-0.17	4	-0.03	5	0.19
P8	8	-0.27	12	-0.04	9	-0.59	9	-0.01
Mean	**8.5**	**-0.21**	**8**	**-0.25**	**10**	**-0.19**	**8**	**-0.14**
SD	**3.38**	**0.25**	**3.74**	**0.17**	**2.97**	**0.27**	**2.36**	**0.23**

Key Points

✓ Stiffness-customized PD-AFOs enable COT reductions

X COT reductions are **not** primarily driven by the paretic ankle

Participants appear to self-organize constituent energetics when wearing a PD-AFO to reduce COT

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Limitations

- Number of participants

Next Steps

- Collect data from more subjects
  - Currently have partial data from 25 subjects
- Does change in COT correspond to a change in muscle activation patterns?
- Are subjects' self-organized changes optimal for reducing COT? Or can training improve their performance?

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Subject	SOC AFO Stiffness (Nm/deg)	PD-AFO Stiffness (Nm/deg)	No AFO Walking Speed (m/s)	SOC AFO Walking Speed (m/s)	PD-AFO Walking Speed (m/s)
1	Hinged AFO	3.10	0.42	0.48	0.58
2	2.55	2.30	0.63	1.02	1.17
3	Solid AFO	3.50	0.36	0.37	0.43
4	Hinged AFO	0.70	0.56	0.63	0.68
5	0.89	0.80	0.78	0.85	0.97
6	0.22	3.60	0.29	0.51	0.62
7	2.12	3.30	0.90	1.11	1.13
8	Hinged AFO	2.40	0.50	0.50	0.84